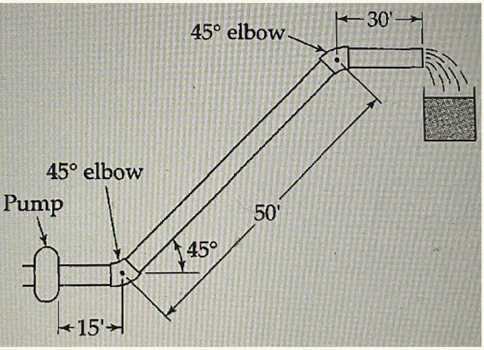
6 A.2



a.) $3.068\text{in} \cdot \dfrac{1\text{m}}{39.37\text{in}} = 0.7793\text{m}$

$w = \left(18 \dfrac{gal}{min} \cdot \dfrac{1min}{60s}\right)\left(3.785 \dfrac{lit}{gal}\right)\left(0.9982 \dfrac{kg}{lit}\right)$

$= 1.13 \text{ kg/s}$

$\langle v \rangle = \dfrac{4w}{\pi D^2 \rho} = \dfrac{(4)(1.13)}{\pi (0.07793)^2 (998.2 \text{ kg/m}^3)} = 0.237 \text{ m/s}$

water $68°F$
height 95ft
diameter 3.068
3in pipe
$\dot{w} = 18 \text{ gal/min}$
$\mu = 1.002 \text{ cp}$
$\rho = 0.9982 \text{ g/mL}$

$Re = \dfrac{D\langle v \rangle \rho}{\mu} = \dfrac{(0.07793m)(0.237 m/s)(998.2 kg/m^3)}{(1.002 \times 10^{-3} \text{ kg/m}\cdot s)}$

$= 1.84 \times 10^4$

Graph $f \approx 0.0066$ solid line

$p_0 - p_L = -\rho g \Delta h + 2 \dfrac{L_e}{D} \cdot \rho \langle v \rangle^2 f$

$= -(998.2)(9.807)\left(-50 \times \dfrac{0.3048}{\sqrt{2}}\right) + 2\left[\left(50 \times \dfrac{12}{3.068}\right) + (2 \times 15)\right] \cdot$

$(998.2)(0.237)^2(0.0066)$

$= 1.055 \times 10^5 \text{ pa} = 15.3 \text{ psi}$

$p_0 - p_L = \left(62.4 \times 0.9982 \dfrac{lbm}{ft^3}\right)\left(32.174 \dfrac{ft}{s^2}\right)\left(\dfrac{50}{\sqrt{2}} ft\right) = 15.3 \text{ psia}$

b.) $\Delta P = \dfrac{(62.4)(0.103)}{144} = 0.0447 / 15.3 = 0.3\%$

6 A.3

Temp $68°F$
length 1320 ft
smooth 6.00 in
pressure diff. 0.25 psi

a.) $p_0 - p_L = \left(0.25 \dfrac{lbf}{in^2}\right)\left(144 \dfrac{in^2}{ft^2}\right)\left(32.174 \dfrac{lbm\, ft}{s^2\, lbf}\right)$

$= 1.15 \times 10^3 \text{ lbm/ft}\cdot s^2$

$\mu = 6.73 \times 10^{-4} \text{ lbm/ft}\cdot s$
$D = 0.5 ft$
$\rho = 62.4 \text{ lbm/ft}^2$
$L = 1320 ft$

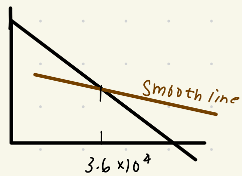
$$Re\sqrt{f} = \frac{D\rho}{\mu}\sqrt{\frac{(p_0 - p_L)D}{2L\rho}} = \frac{(0.5)(62.4)}{6.73\times10^{-4}}\sqrt{\frac{(1.15\times10^3)(0.5)}{2(1320)(62.4)}} = 2.74\times10^3$$

b.) $Re\sqrt{f} = 2.74\times10^3$ $\quad f = \left(\frac{2.74\times10^3}{Re}\right)^2$

$$Re = \frac{2.74\times10^3}{\sqrt{1}} = 2740$$ when f is 1 Re is 2.74×10^3

$$Re = \frac{2.74\times10^3}{\sqrt{0.01}} = 2.74\times10^4$$ when f is 0.01 Re is 2.74×10^3

plot points in graph



$$\langle v\rangle = \frac{Re}{(D\rho/\mu)} = \frac{3.6\times10^4}{4.64\times10^4} = 0.78\ ft/s$$

$$\dot{m} = \frac{\pi D^2}{4}\langle v\rangle = \frac{\pi\cdot(0.5)^2}{4}(0.78) = 0.152\ ft^3/s = 68\ \text{us gal./hr}$$

6 A.9

$D_t = 4in$

Length $= 6.5ft$

$D_g = 1/16in$

void fraction $= 0.41$

Temp $= 300K$

$\mu = 1.495\times10^{-4}\ g/cm\cdot s$

$P_0 = 25atm$

$P_L = 3atm$

$$\rho = \frac{PM}{RT}$$

$$-\frac{dp}{dz} = 150\frac{\mu G_0}{\rho D_p^2}\frac{(1-\varepsilon)^2}{\varepsilon^3} + \frac{7}{4}\frac{G_0^2}{\rho D_p}\frac{(1-\varepsilon)}{\varepsilon^3}$$

integrate $z=0$ to $z=L$

$$\left[150\frac{\mu G_0}{D_p^2}\frac{(1-\varepsilon)^2}{\varepsilon^3} + \frac{7}{4}\frac{G_0^2}{D_p}\frac{(1-\varepsilon)}{\varepsilon^3}\right]L = \frac{M}{2RT}(p_0^2 - p_L^2)$$

$$150\frac{(1.495\times10^{-4})(G_0)}{2.54/16}\frac{(1-0.41)^2}{(0.41)^3}(5.5\times30.48) + \frac{7}{4}\frac{(G_0)^2}{2.54/16}\frac{(1-0.41)}{(0.41)^3}(5.5\times30.48)$$

$$= \frac{44.01}{2(8.31451\times10^7\times300)}\left[(25(1.013\times10^6)\times10^7)^2 - (3.0399\times10^6)^2\right]$$

$15820\ G_0^2 + 753.4\ G_0 = 5.58\times10^5$ $\longrightarrow$ find roots

± 5.92

$$\dot{w} = \left(\frac{\pi}{4}\right)D^2 G_0 = (0.7854)(4\times2.54)^2(5.92) = 480\ g/s$$

6B.2

a.) Eq 4.4-30 $F_K = 1.328\sqrt{\rho \mu L W^2 v_\infty^3}$

$Re_L = \dfrac{L V_\infty \rho}{\mu}$ $F = AKf$

$$f = \frac{F_K}{AK} = \frac{1.328\sqrt{\rho \mu L W^2 V_\infty^3}}{2WL\frac{1}{2}\rho V_\infty^2} = 1.328\sqrt{\frac{\mu}{LV_\infty \rho}} = \frac{1.328}{\sqrt{Re_L}}$$

b.) $F_K = 0.072\rho V_\infty^2 WL(L_\infty \rho/\mu)^{-1/5}$

$$f = \frac{F_K}{AK} = \frac{0.074\rho V_\infty^2 WL(LV_\infty \rho/\mu)^{-1/5}}{2WL\frac{1}{2}\rho V_\infty^2} = \frac{0.074}{Re_L^{1/5}}$$

6B.3

a.) $w = \dfrac{2}{3}\dfrac{(\mathcal{P}_0-\mathcal{P}_L)B^3 W\rho}{\mu L} = \rho\langle v_z\rangle(2BW)$ $F_K = (\mathcal{P}_0-\mathcal{P}_L)(2BW)$

$$f = \frac{F_K}{AK} = \frac{(\mathcal{P}_0-\mathcal{P}_L)(2BW)}{(2WL)(\frac{1}{2}\rho\langle v_z\rangle^2)} \Rightarrow \frac{(\mathcal{P}_0-\mathcal{P}_L)(2BW)}{(2WL)(\frac{1}{2}\rho\langle v_z\rangle)} \cdot \frac{3\mu L}{(\mathcal{P}_0-\mathcal{P}_L)B^2} = \frac{12}{2B\langle v_z\rangle\rho/\mu} = \frac{12}{Re}$$

b.) $R_h = \dfrac{S}{Z} = \dfrac{2BW}{2W+4B} = B$

$$f = \frac{16}{D\langle v_z\rangle\rho/\mu} = \frac{16}{4R_h\langle v_z\rangle\rho/\mu} = \frac{16}{4B\langle v_z\rangle\rho/\mu} = \frac{16}{4B\langle v_z\rangle\rho/\mu} = \frac{8}{2B\langle v_z\rangle\rho/\mu} = \frac{8}{Re}$$

6B.5

a.) $f = \dfrac{0.0791}{Re^{1/4}}$ $F_K = (\mathcal{P}_0-\mathcal{P}_L)\pi R^2$ 6.1-2 $F_K = (2\pi RL)(\frac{1}{2}\rho\langle v\rangle^2)f$

$(\mathcal{P}_0-\mathcal{P}_L)\pi R^2 = (2\pi RL)(\frac{1}{2}\rho\langle v\rangle^2)f \Rightarrow (\mathcal{P}_0-\mathcal{P}_L) = 2\frac{L}{R}(\frac{1}{2}\rho v^2)f$

$R = D/2$

$\dfrac{L}{R} = \dfrac{L}{D/2} = 2\dfrac{L}{D}$

$(\mathcal{P}_0-\mathcal{P}_L) = 2(2\frac{L}{D})(\frac{1}{2}\rho v^2)f = 2\frac{L}{D}\rho v^2 f$

$Re = \dfrac{\rho\langle v\rangle D}{\mu} = \dfrac{\rho D}{\mu}\left(\dfrac{4w}{\pi\rho D^2}\right) = \dfrac{4w}{\pi D\mu}$ $Re \propto \dfrac{w}{D}$

$w = \rho VA = \rho V\dfrac{\pi D^2}{4} \Rightarrow 4w = \rho V\pi D^2 \Rightarrow V = \dfrac{4w}{\pi\rho D^2}$ $v^2 \propto \dfrac{w^2}{D^4}$

$$f \propto Re^{-1/4} \propto \left(\frac{w}{D}\right)^{-1/4} = w^{-1/4} D^{1/4}$$

$$(P_0 - P_L) \propto \frac{L}{D} \rho v^2 f \propto \frac{1}{D} \cdot \frac{w^2}{D^4} \left(w^{-1/4} D^{1/4}\right) \propto D^{-1-4+1/4} = D^{-19/4}$$

$$\frac{(P_0 - P_L)_{new}}{(P_0 - P_L)_{old}} = \left(\frac{D/2}{D}\right)^{-19/4} = \left(\frac{1}{2}\right)^{-19/4} = 27$$

b.) $Re = \frac{4w}{\pi \mu D} \Rightarrow$ than use moody chart where we get f & $Re \Rightarrow$ from $\frac{k}{D}$

$D_2 = \frac{D}{2}$ $Re = \frac{4w}{\pi \mu D_2} \Rightarrow$ Do same as above $\uparrow$ $\Rightarrow \frac{P_2}{P_1} = \frac{f_2}{f_1} \left(\frac{D_2}{D}\right)^{-5}$

6 B.7

$$F_z = m \frac{dv}{dt} \qquad F_z = mg_z - F_{kz} = mg - (\pi R^2)\left(\tfrac{1}{2}\rho v^2\right)(0.44) = 0.22 \pi R^2 \rho v^2$$

$$m \frac{dv}{dt} = mg(1 - c^2 v^2) \qquad \frac{dv}{dt} = g(1 - c^2 v^2)$$

$v(0)=0$
$v(t)=0$

$$\int_0^v \frac{dv}{1 - c^2 v^2} = g \int_0^t g\,dt \qquad \int_0^v \frac{dv}{1 - c^2 v^2} = \frac{1}{c} \tanh^{-1}(cv) \Rightarrow$$

$$\frac{1}{c} \tanh^{-1}(cv) = gt \Rightarrow \tanh^{-1}(cv) = cgt$$

$$\frac{dz}{dt} = v = \frac{1}{c} \tanh cgt$$

$$z(t) = \int_0^z dz = \frac{1}{c} \int_0^t \tanh cgt\,dt \Rightarrow z = \frac{1}{c^2 g} \ln \cosh cgt$$

b.) $v(t \to \infty) = \frac{1}{c} \tanh(cgt) = \frac{1}{c}$

$$m = \frac{4}{3} \pi R^3 \rho_{sph} \qquad c^2 = \frac{0.22 \pi R^2 \rho}{\left(\frac{4}{3} \pi R^3 \rho_{sph}\right) g} = \frac{0.22 \rho}{\frac{4}{3} R \rho_{sph} g} = \frac{0.165 \rho}{R \rho_{sph} g} \Rightarrow c = \sqrt{\frac{0.165 \rho}{R \rho_{sph} g}}$$

$$v(t \to \infty) = \sqrt{\frac{R \rho_{sph} g}{0.165 \rho}}$$